Chapter 3.

THE CAUSES OF HAPPINESS AND MISERY

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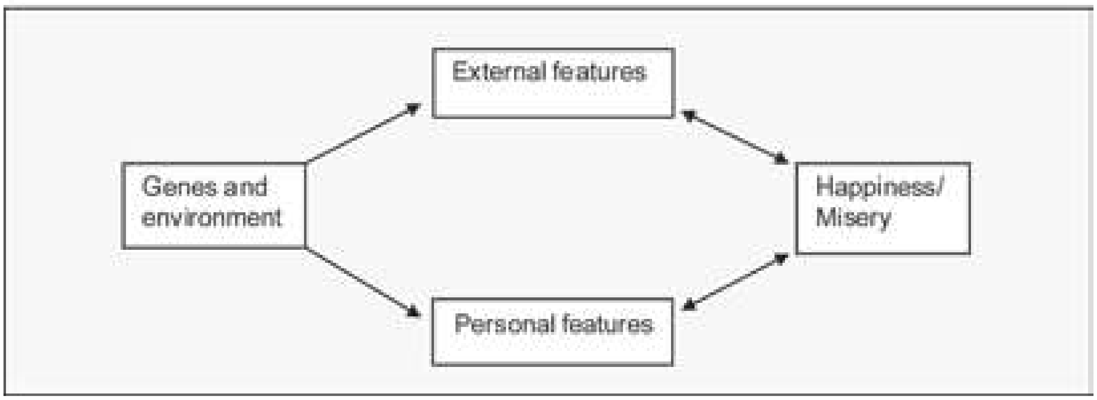
If we want to influence the levels of happiness and misery, we need to know what causes them. As a result of some 30 years of research, we now know a good deal about this. [1]

If we think of each individual, every one of us has her own genetic make-up, but the person she becomes depends on the interaction of those genes with the environment she encounters. Together, genes and environment determine the main features of a person's life – both those that are very "personal" and those that are more clearly "external." And these features in turn determine a person's well-being, as illustrated in Figure 3.1.

Figure 3.1



Among the more "external" factors, key determinants of happiness include:

- income
- work
- community and governance, and
- values and religion

and, among the more "personal" features, key determinants include:

- mental health
- physical health
- family experience
- education, and
- gender and age

Thus a person's happiness at a point in time is determined by the whole of her life course. The current external features of her life are important, but so are the personal features that have developed over the previous course of her life.

For policy-makers the main issues are the environmental factors affecting happiness, since these are what can be changed. Ideally we would study their effects holding the genes constant, but most research so far has been unable to do this.[2] So in this chapter we look directly at the ways in which external and personal factors (however they arise) affect a person's happiness.

What the policy-maker wants to know is how big an effect each factor has on happiness. So we concentrate both on that, and on the share of variance explained by the factor (which depends of course on the sample of people being considered).

We look first at the more external factors (income, work, community, governance, values and religion) and then at the more personal ones (mental health, physical health, family, education, gender and age). In many cases there is a two-way interaction between the factor and happiness. For example, education affects a person's happiness, but happiness also affects the ability to learn. Likewise health affects happiness and happiness affects health. That is why, in the diagram above, arrows run from personal and external factors towards happiness (one direction of causality) but also run from happiness to health, education and so on (reverse causality). In what follows, we discuss each individual factor one by one, including (where it exists) the two-way workings of causality.

Types of Evidence

On every factor we offer a wide range of evidence. To isolate the causal effect of each factor is not easy. It clearly requires us to hold as much else as possible constant while we look at the co-movement of well-being and the factor in question. In most cases this cannot be done experimentally. So the next best is to study the same individuals (or countries) over time and see how their well-being moves when different factors change. Much of the evidence we shall quote is of this longitudinal, time-series form.

But some insights can also be got from cross-sectional evidence. In this case we are comparing different individuals (or countries) at the same point in time. The problem here is that, when we compare individuals or countries, there are many ways in which they may differ (for example in personality or values) that cannot easily be measured and controlled for when we are examining the effect of those factors that can be measured. But when we have longitudinal data on the same person or the same country we can assume that these unmeasured factors are more similar at each observation, and may have a better chance of tying down what is causing what.[3]

As a background to what follows we include in Appendix A standard individual happiness equations using two well-known sets of panel data and the World Values Survey. We give both cross-sectional and longitudinal estimates.

We can now review the main causes of happiness one by one.[4] We begin with the more "external" causes.

Income

Does economic growth improve the human lot? In 1974 Richard Easterlin wrote a seminal article on what has become known as the Easterlin paradox. He presented evidence of two apparently contradictory phenomena.

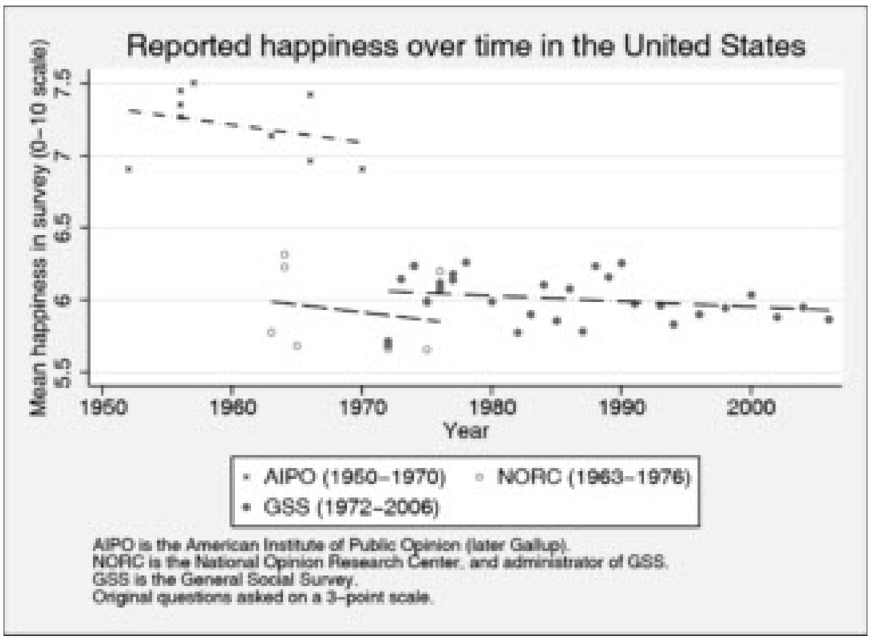
> "Fact 1" At a point in time within any society, richer people are on average happier than poorer people (a cross-sectional "fact").
>
> "Fact 2" Over time within many societies, the population does not on average become happier when the country's income rises (a time-series "fact").

To reconcile Facts No 1 and 2 Easterlin proposed the relative income hypothesis. People are comparing themselves with other people: it is relative income rather than absolute income that matters. Thus at any particular point in time richer people would compare favorably with poorer people (explaining "Fact No 1"). But over time, the aggregate of relative income in the population remains constant (thusexplaining "Fact No 2").

There is no doubt that Fact No 1 is correct. In multiple regressions, income always emerges as a factor explaining the variation in life satisfaction within a country – not the most important factor (see below) but an important one. It is also now possible to tie down quite closely the functional form of that relationship. It is well described by a logarithmic form where the absolute level of life satisfaction varies linearly with the logarithm of income.[5] This means that, for example, an extra dollar increases the satisfaction of a poor person by 10 times as much as it increases the satisfaction of a person who is 10 times richer. For centuries people have intuitively believed in the "diminishing marginal utility of income" but happiness research now provides empirical estimates that policy-makers can use when considering the distributional impact of their policies.

Turning to the time-series Fact No 2, which is what is relevant to the policy debate, the state of research is more unsettled. From micro-economic evidence presented below we know that relative income does matter. This is also supported by experiments in neuroscience. This establishes a strong prior that the time-series effects of higher absolute income would be less than those found in cross-sectional studies when the average level of income is held constant. This prior is in turn reinforced by the finding that in some important countries, especially the United States, average happiness has not risen despite strong economic growth (see Figure 3.2).[6] This was so during the golden period of economic growth in the 1950s and 1960s, and also more recently, when even the top quintile of income recipients experienced no growth in happiness, despite huge increases in their income.[7]

Figure 3.2



However the experience of particular countries is not enough to support generalized statements about the relation between happiness and economic growth. That requires a more exhaustive study of as many countries as possible. We shall therefore proceed as follows. First we shall examine the micro-economic evidence about how relative and absolute income influences the happiness of individuals. This will also include a discussion of adaptation. That done, we shall turn to aggregate evidence about the experience of whole nations.

Individual happiness and income

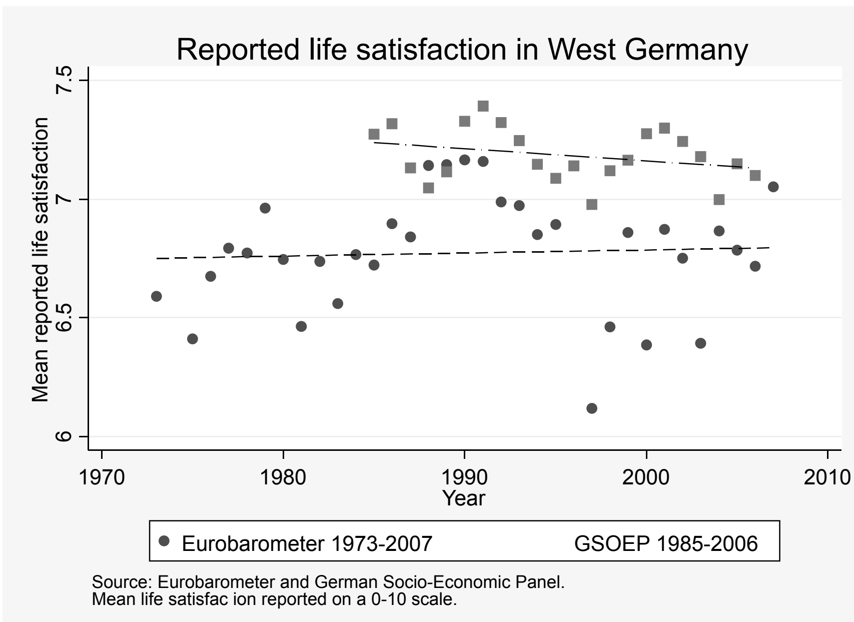
Does relative income raise a person's happiness and does absolute income do likewise? To examine the effect of income on happiness, we must eliminate any effect of a person's underlying happiness upon their income. The best way to attempt this is with panel data in which we trace the same individual over many years and examine how changes in the person's income affects their subsequent happiness. Fortunately we have such data from Europe's leading country. In West Germany the German Socio-Economic Panel (GSOEP) has been tracking the same individuals each year since 1984. We can use these data to help us understand the movement of average life satisfaction in that country, as illustrated in Figure 3.3, using the Eurobarometer series since 1972 and the GSOEP since 1984.

The first finding is about the effect of own income on life satisfaction: ceteris paribus, differences in income explain about 1% of the variance of life-satisfaction in the population. It is a significant effect, though when we use the panel feature of the data, it is somewhat reduced.[8]

The next step is to use the panel data to decompose this effect of income into an effect of absolute income and an effect of income relative to the appropriate comparator group. For this purpose relative income is measured relative to other people of the same sex, age and education in the year in question. When this analysis is performed with suitable controls, there is no effect left for absolute income.[9] Only relative income matters and this is clearly what explains the fact that in Figure 3.3 average life satisfaction has not risen despite rapid economic growth.

A third result is also of interest. Many people, including some psychologists, use adaptation to explain why happiness is not permanently increased by higher income: "whatever your income, you get used to it." This explanation clearly has problems since, if it were wholly true, we should not observe Fact No 1.[10] And in the GSOEP data there is no strong effect on current life satisfaction of current income relative to income over the previous three years – no evidence, that is, of a role for adaptation.[11]

Figure 3.3



We can turn now to the dozens of cross-sectional studies which also indicate strong effects of income comparisons.[12] All cross-section studies run the risk of exaggerating the effect of income on happiness by including the reverse effects of happiness on income.[13] But many of them provide more useful detail, including explicit questions about whom people compare themselves with, and how they think their income compares with that of different groups. For example, in a representative sample of rural Chinese, people said they mainly compared themselves with others in the same village, and multiple regression results showed that among all possible factors the most important for happiness was perceived relative income within the village.[14]

In advanced countries the comparators are different. The European Social Survey asked people "How important is it for you to compare your income with other people's incomes?" and those who said income comparisons were more important were also on average less satisfied with their lives – a common finding.[15] Respondents were also asked "Whose income would you be most likely to compare your own with?" The most important group mentioned was "colleagues," and the same was found in a one-year-only set of questions in the GSOEP.[16] In the GSOEP study people were then asked to rank their income compared with their colleagues, and also with their friends, neighbors, etc. In explaining life satisfaction it was confirmed that perceived relative income has a large effect on life satisfaction. Similar findings have been found in repeated cross-sections in the United States,[17] which helps to explain the fact that happiness has not increased in the U.S. (see Figure 3.2).

The preceding studies use data on perceived relative income. This perception can of course be influenced by the mood of the respondent, but it is reassuring that the results are very similar if we only include actual relative income. Many studies only do this, and most but not all of them find significant effects of income relative to income in the surrounding area, be it travel-to-work area, county or province.[18]

If happiness depends on income relative to the income of a "comparator" that means that when the "comparator's" income rises, happiness falls. As we have seen, this is the most common general finding, but it is not always the case. In some cases people appear to take the comparator's income as an indication of what they might themselves attain, rather than as an external benchmark they need to compare well with. The comparator's income is then like a light at the end of the tunnel.[19] In a number of situations this "tunnel effect" has been found to dominate the "external norm effect,"[20] so that comparator income sometimes has a positive effect or in other studies a zero effect.[21]

But the more general finding is that comparator's income reduces happiness and this has been strikingly confirmed in many laboratory experiments. One neuroscience experiment involved the task of guessing the number of dots on a screen.[22] Good guesses were rewarded by a monetary payment. Each subject was paired with another subject, and after each of the 300 trials the subject was told the accuracy of his own guesses and the associated income he would receive, as well as the same information for his "pair." At the same time fMRI scans measured the blood oxygenation in the subject's relevant reward center (the ventral striatum). Blood oxygenation responded strongly to both the subject's own income (positively) and to the pair's income (negatively). And the negative affect of the pair's income was at least two thirds as large as the positive effect of the subject's own income.

Country-level income and happiness: cross-sectional evidence

We can turn now to country-level data and ask, how far does absolute income affect the happiness of nations? We can begin with a cross-sectional analysis of data similar to those presented in Chapter 2, before turning to the time-series, which is far more informative about what causes what.

In Table 3.1, we examine the Gallup World Poll's measures of well-being reported in Chapter 2. We show first a simple regression of average well-being (according to each measure) on the log of GDP per head across the sample of countries. There is a very strong relation.[23] However material well-being is not the only determinant of well-being. In order to get a sensible idea of its effect, we should also include other obvious determinants. Beginning with the other indicators in the UNDP's Human Development Index (HDI), we include:

Health (we use healthy life expectancy from the WHO)
Education (we use the HDI average educational level among adults)

We also include measures of the degree of social support, freedom and corruption that individuals experience in their country. These come from the World Gallup Poll's answers to the following questions:

Social Support

If you were in trouble, do you have relatives or friends you can count on to help you whenever you need them, or not? (proportion of respondents saying Yes)

Freedom

In your country, are you satisfied or dissatisfied with your freedom to choose what you do with your life? (proportion of respondents saying Yes)

Corruption

Average proportion of respondents saying Yes to the following 2 questions:
1. Is corruption widespread within business located in your country, or not?
2. Is corruption widespread throughout the government in your country, or not?

Finally we take into account the strength of family life – measured by the proportion of people separated, divorced, or widowed.

In Panel A of Table 3.1 we include only income as an explanatory factor. It has a strong positive impact on life evaluation, a smaller impact on positive affect, and an insignificant impact on negative affect. For life satisfaction the β-coefficient on income is high at 0.81; it thus explains 65% (β2) of the variation across countries. However, when in Panel B we introduce the social variables discussed above, the positive effect of income falls sharply – by more than half. Most of the social variables are highly significant. When it comes to positive and negative affect, only the social variables play a significant role.

A parallel analysis focusing only on European countries shows similar results using the European Social Survey. The dependent variable is the average of life satisfaction and happiness these days. When regressed on log GDP per head only, β is .84. but when we introduce one additional variable – the average of social trust and trust in police -the β-coefficient on trust is .62, and that on GDP falls to .36.

Table 3.1 Regressions to explain average well-being across countries[24] (standardized βstatistics)

	Dependent Variable					
Independent Variables	Life evaluation		Positive affect		Negative affect	
Panel A						
Log GDP per head	0.81	***	0.40	***	-0.08	
$\bar{R}^2$	0.65		0.15		-0.00	
No of countries	153		153		153	
Panel B						
Log GDP per head	0.28	**	-0.18		0.22	
Health	0.25	**	0.24		0.27	
Education	-0.01		-0.18		-0.05	
Social support	0.29	***	0.43	***	-0.35	***
Freedom	0.15	***	0.49	***	-0.24	**
Corruption	-0.18	***	0.00		0.23	***
Divorce etc.	-0.43		-0.09		-0.08	
$\bar{R}^2$	0.80		0.52		0.20	
No of countries	139		139		139	

Significance Levels: (1 tailed tests)
* 0.05 ** 0.01 *** 0.001

The preceding analyses underline the problems of studying the relation of national income and happiness without taking into account other variables. This is the main problem with the careful study by Betsey Stevenson and Justin Wolfers in which they compare the effect of income on life evaluation at the cross-country level with its effect at the individual level within a country.[25] They argue that there can be no effects of comparator income at the individual level if (as they find) the cross-country effects are as high as the within-country individual effects. This statement is logically correct, provided other things are held equal. But they are not: the "effect of income" at the cross-country level is estimated with nothing else held constant. But, as we have shown in Table 3.1, the cross-country effect falls sharply when other variables are included. It is of course possible that high income in a country is good for health, social support, freedom and corruption. But to find out about the direct effects of comparator income on family well-being, we would definitely have to keep these other things constant. Moreover from a public policy point of view it is important to separate out the effects of income from those of health, social support, freedom and corruption, and not to roll them all together.

The problem of other things equal maybe relatively less acute when we now turn to changes in GDP over time (not holding other things constant).

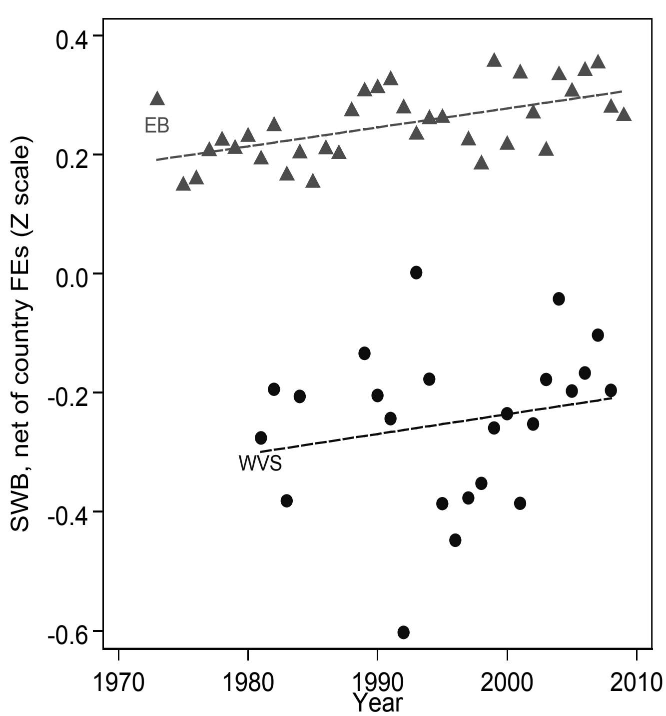
Country-level income and happiness: long-term economic growth

So we are now ready to scrutinize Richard Easterlin's Fact No 2. Is it true that, if countries grow in income, they become no happier? On this matter, one can make no general statement. As we have seen there are some countries like the U.S. and West Germany that have grown over long periods of time but have not become happier.[26] On the other hand there are other countries where income growth has gone hand in hand with increases in happiness.

It would however be helpful if there were some way to summarize the average of this relationship across all countries. In a recent paper, Richard Easterlin offered his own summary.[27] In order to concentrate on long-term economic change, he confined himself to 37 countries with a long enough range of data (21 years for developed countries, 15 for developing countries and 12 for transition countries). He found in each group a flat or negative relation between changes in life satisfaction and income per head.

However, this analysis has been powerfully challenged by Stevenson and Wolfers, whose most recent paper deals entirely with the issue of long-term growth. It shows first that, both in the countries covered by the World Values Survey and in those covered by Eurobarometer, there has been an increase in life satisfaction in the average country over recent decades, see Figure 3.4.[28] They then investigate whether the intercountry differences in changes in life satisfaction are associated with different rates of growth of per capita income. For the World Values Survey they find a strong relation between changes in life satisfaction and changes in trend-GDP per head – roughly equal in size to the effect found in simple cross-country regressions. They also find a relationship among Eurobarometer countries, though the size of the effect is under half of that in the cross-section. One reason why their findings differ from Easterlin's is that they exclude all countries for which the survey did not cover the whole country.[29]

Figure 3.4 Average subjective well-being in countries covered by Eurobarometer (EB) and World Values Survey (WVS)



There are clearly a number of issues remaining to be resolved in this area.[30] But a reasonable interim conclusion is as follows:

1. In a typical country, economic growth improves happiness, other things equal. But other things are not necessarily equal, so economic growth does not automatically go with increased happiness. Thus policy-makers should balance the argument for more rapid growth against the arguments for supporting other sources of happiness. This applies to countries at every level of development.

2. In developed countries in particular there is strong micro-level evidence of the importance of income comparisons, which has not been disproved by aggregate data. For this reason policies to raise average happiness must target much else besides economic growth.

Country-level income and happiness: cyclical fluctuations

There is of course a sharp distinction between long-term economic growth, which may have little effect on the level of unemployment, and short-term growth, which is the only way to reduce the high unemployment currently prevailing in most parts of the world.

Everybody agrees on the importance of short-run growth in such a context. Happiness fluctuates over the business cycle. It is generally somewhat higher when employment is high relative to trend and when unemployment is therefore low.[31] But on the other hand happiness is also lower when inflation is high, as often happens in upturns. In the overall balance, happiness rises in booms because a one-point decrease in unemployment has at least twice as large an effect on happiness as a one-point increase in the inflation rate.

Economic stability is a crucial goal for any society, due largely to the fact of loss aversion, whereby individuals hate to lose x dollars more than they love to gain x dollars.[32] But economic stability is a quite different goal from long-term economic growth. Long-term growth has much less impact on human happiness than do human relationships in all their dimensions – as we shall see.

Work

A key relationship comes through work. It provides not only a livelihood but a source of meaning – feeling needed and able to contribute. But not everyone can get work, nor if they can, is it always satisfying.

Unemployment

When people become unemployed they experience sharp falls in well-being and their well-being remains at this lower level until they are re-employed.[33] The estimated effect is typically as large as the effect of bereavement or separation, and the unemployed share with these other experiences the characteristic of ceasing to be needed.

The Appendix to this chapter documents that unemployment reduces well-being in all the datasets analyzed. It also shows that the main impact of unemployment on well-being is not through the loss of income, but rather through loss of social status, self-esteem, workplace social life, and other factors that matter.

Psychologists[34] have examined these non-pecuniary benefits of work, and they include the preset time structure of the working day, regularly shared experiences and contacts with people outside the family, links to goals and purposes that transcend the individual, personal status and identity, and the enforcement of activity.[35] Unemployment is destructive due to its negative effect on these functions.

High unemployment also has spillover effects not only on the families of the unemployed[36] but also on those in work, who feel less secure in their jobs.[37] Thus private sector employees are more affected than public sector employees, whose jobs are more secure.[38] When we total up all the well-being effects of a rise in the unemployment rate, the loss to the rest of the population (which is a large number of people) is twice as large as the loss to the unemployed themselves.[39]

There is also another spillover effect of the unemployment rate: upon the unemployed themselves. A priori one might expect individual unemployed people to be worse off if more other unemployed people were competing with them for the available jobs. However, the evidence suggests that greater unemployment may actually reduce the stigma associated with one's own unemployment, or be associated with greater social support.[40] British data reveals that others' unemployment (at the regional, household, and couple level) generally has a positive effect on the well-being of the unemployed (at least for men).[41] This social norm effect, whereby my own unemployment hurts less when more other people are unemployed, parallels data on suicide and para-suicide by the unemployed, which is more probable where unemployment is low.[42]

But if unemployment is bad, is any reduction in it unambiguously good? In particular is it better to get people into bad jobs rather than no jobs at all? This has been a subject of controversy, but a comprehensive study using the German Socio-Economic Panel concludes that "Our main result is that we cannot identify a single job feature, nor a combination of such features that constitute such low quality jobs that remaining unemployed would be the better choice for the individual. On the contrary, the bulk of our evidence shows that even low quality jobs are associated with higher life satisfaction, and this effect is statistically significant for most specifications of "bad" jobs.[43] A parallel study examines the value of the large German workfare program and concludes that people's life satisfaction rises substantially after moving onto the program from being totally out of work.[44]

Quality of work

Thus one of the most important aspects of the labor market in terms of well-being is whether individuals are able to find a job, given that they want one. However, when in work the quality of life at work is also crucial. The view that job quality consists of pay and hours of work has by now largely been superseded.[45] In three waves of the International Social Survey Programme workers rank eight different job characteristics, on a one to five scale from "Not at all important" to "Very Important." The characteristics are: high income, flexible working hours, good opportunities for advancement, job security, interesting job, allows to work independently, allows to help other people, and useful to society. The results[46] show that only around 20% of respondents in OECD countries say that having a high income is very important and the same figure applies to flexible hours and promotion opportunities. But around 60% say that job security is very important, with similar figures for interesting work and autonomy (50% and 30% respectively).

Thus it is not surprising that measured satisfaction is shown to be strongly correlated with not only pay at work, but also measures of job security, autonomy, workplace trust, independence and so on. Any evaluation based solely on income and hours will omit many key characteristics that workers value.

An important school of thought focuses on the importance of intrinsic motivation at work – and related to that the importance of the intrinsic features of the job (rather than pay) as sources of satisfaction:[47] eudaimonic returns associated with human flourishing. These features include a sense of overall purpose for the job, a degree of autonomy in discharging it, and the competence to do the job – a proper fit between worker and job. Allied to this people need support and recognition for their efforts. Experimental work has underlined the role of purpose and control in determining individual behavior in hypothetical-choice experiments.[48]

This approach has a number of implications. It plays down the role of pay as the prime system of motivation. Psychologists have shown many cases where introducing financial incentives reduces performance by undermining intrinsic motivation.[49] These findings need to be taken seriously by those who design systems of performance-related pay.

One striking finding of happiness research is that the time of day when people are least happy is when they are in the presence of their line manager.[50] This suggests that too many managers fail to inspire their workers and rely too much on mechanical incentives and command.

Worker well-being matters to firms as well as workers; it is a good predictor of productivity. It is well-known that workers who are more satisfied with their jobs are less likely to quit their jobs. They are also less likely to reduce firm productivity via absenteeism or via presenteeism - turning up to work, but contributing little.[51]

Self-employment

When it comes to autonomy, some workers can completely control their quality of work, because they are self-employed. The self-employed do worse on many job dimensions, including income, hours of work and job security, but even so they often report higher levels of overall job satisfaction than do the employed, at least in OECD countries. A positive correlation is found in American and European data,[52] and in data from Great Britain, Germany and Switzerland.[53]

This need not necessarily mean that the self-employed experience greater overall satisfaction with their lives, if they are sacrificing other dimensions of their lives to their job. Thus it is interesting that in Appendix A, Table 2, self-employment has no significant effect on overall life satisfaction.

The great majority of work on well-being and self-employment has used OECD data. But any comparison of self-employment and employment will depend critically on the extent to which the former is a choice.[54] If individuals are free to choose, they will choose self-employment if their happiness there is higher. However, when there are insufficient employment opportunities in the formal sector, self-employment may not be a choice but a necessity. If formal-sector employment opportunities are positively linked to financial development, this could help to explain a more positive job satisfaction gap between the self-employed and the employed in more developed countries. This proposition has been tested on data from the World Value Surveys (WVS) over the 1981-2001 period.[55] The self-employed do not always report higher job satisfaction scores than employees, but do so more often in developed countries. This pattern is not affected by the inclusion of income as a control variable, suggesting that the key difference between employees and the self-employed is to be found in the non-pecuniary arena.[56] The results for non-OECD countries are line with the fact that self-employment is associated with lower satisfaction in Latin American countries.[57]

Retirement

Eventually many of us stop working: we retire. Do people enjoy life more after they retire? Analyses of panel data from the Survey of Health, Ageing and Retirement in Europe (SHARE) show no large overall effect but a wide disparity in the effect on different individuals: more educated workers experience rises in well-being on retiring; others' well-being falls.[58] So workers in lower-quality jobs are not necessarily those who gain most from retirement. The lower-educated report lower levels of well-being when in work, as might be imagined, but they also report a greater drop in well-being when they retire. This seeming paradox might be explained if education affects not only the value of employment, but also that of retirement: the lower-educated may have worse jobs than the better-educated, but also worse retirements. However most work in this area is still preliminary.

It might be expected that retiring voluntarily would be associated with a greater rise in well-being than involuntary retirement. However, it is not always easy to distinguish the type of retirement in survey data. Also, those who retire voluntarily may have a reason for doing so, such as being in poor health, which will potentially confound any comparison of voluntary and involuntary retirement if it is not controlled for.

Social Capital

While work is one important part of our social world, our relationships go much further, and include relationships with family, friends and community. In Maslow's pyramid of human needs, love and belonging come just after basic physiological and safety needs.[59] Clearly, the sources of individual happiness include the set of social interactions through which individuals are interconnected. The quantity and quality of social relations in a community is sometimes referred to as social capital.[60]

Why "capital"? Because people's social networks are accumulated over time (like financial capital) and because they yield benefits, such as informal mutual assistance or simply the pleasure of being socially integrated and participating in intense social interactions. As a network, social capital also includes a notion of externality, i.e. mutually reinforcing benefits for all members.

We have already seen in Table 3.1 how strongly the happiness of nations is influenced by the extent to which the citizens believe they have others they can rely on in times of trouble. We have also seen the powerful effect of the levels of perceived corruption in government.

Trust

In a well-functioning society there is a high level of trust – above all between citizens, but also in institutions. There is a standard question that has been asked in many surveys over many years in many countries. "In general, do you think that most people can be trusted, or, alternatively that you can't be too careful in dealing with people?"

This has been asked in fewer countries than the questions used in Table 3.1. But among those where it has been asked, it performs at least as well as the social variables in Table 3.1. One might ask, Do answers to these questions correspond to real differences between countries? Their validity is confirmed by the "lost wallet" experiment, first conducted by the Reader's Digest Europe in 1996. This experiment involved dropping 10 cash-bearing wallets (including a name and address) in each of 20 cities in 14 western European countries, and in each of a dozen U.S. cities. Researchers[61] later used these data to validate the classic question of trust. It turned out that indeed, the actual frequency of return of the wallets was highly correlated with national average social trust, as measured in international surveys. Since then, this experiment has often been replicated, and the question about the likelihood that a lost wallet, if found by a stranger, or alternatively by a police officer, would be returned intact to the owner was used in the Gallup World Poll, as well as certain national surveys (e.g. in Canada and the United States) to provide more specific measures of trust.[62]

Studies of various types of trust important to Canadian individuals showed large effects for trust in neighbors, trust in police, trust in strangers, and especially workplace trust.[63] Higher life satisfaction is correlated with having a more intense relational life in general, such as socializing frequently with friends and relatives,[64] attending social gatherings and cultural events, participating in sports, performing volunteer work,[65] and pro-social behavior (donations of time, donations of money, providing help to a stranger). Such correlation may include an element of reverse causality with happier people more likely to enter these situations. But several studies have documented the stability of trust over generations: the social trust of descendants of immigrants to the United States[66] or Canada[67] is positively linked with the trust level of their ancestors' home-country. The current level of trust in Europe[68] and Africa[69] can be traced back to distant past "critical junctures" such as slavery or other historical conditions. These studies suggest that trust causes life satisfaction rather than the reverse.

Levels of trust have fallen substantially over time in some countries (like the U.S. and U.K.) and risen in others (such as Denmark and Italy). This may help to explain the fact that life satisfaction has not risen in the U.S. and U.K., while it has risen in a number of continental European countries. Indeed for the U.S. it has been well argued that the main offsets to the private benefits of economic growth include not only comparator incomes but also a decline in the quality of human relationships, as measured by increased solitude, communication difficulties, fear, distrust, family infidelity and reduced social engagement.[70]

Bonding and bridging capital

At this point it is important to contrast the relations between people who are similar to each other (bonding capital) with the relations between people who are different (bridging capital).[71] We want both - not only good social capital within communities but also good links between communities.

The first is the more obvious. Social capital has a local dimension and is most evident within communities, i.e. sub-groups of the population who interact directly and frequently share common norms[72] and a sense of common identity. Staying rooted in the same neighborhood for a longer time is associated with higher levels of all types of trust, especially neighborhood trust;[73] while respondents who live in districts where the population is highly mobile are less likely to trust their neighbors.[74] But longer-distance attachments to similar people also often matter a lot, even if not as much as local attachments.

So may more social capital for some mean less for others, because they are in some way excluded? Studies of migrants confirm the importance of close communities. New migrants are often found to be less satisfied with their life than natives, even when they share an identical socio-professional situation. This is certainly related to the fact that migrants have to leave behind them their networks of friends and family.[75] It could also be due to racial discrimination, although racial tolerance has increased in many Western countries over the last decade, leading to an improvement in the happiness of minorities.[76] This bridging capital implies that typical communities now involve widened circles, so that one can now identify with a wider range of people.[77]

Freedom

Another key feature of a society is the freedom that it provides to its members. No people can be truly happy if they do not feel that they are choosing the course of their own life – subject always of course to the inevitable constraints of human existence. The importance of freedom is confirmed in Table 3.1.

It is also the fact that the least happy societies documented in 1990 were those in the former Soviet bloc. It is not easy to disentangle the effect of the transition, but earlier surveys for Hungary and for Tambov district in Russia also show low levels of happiness at that level of GDP per head.[78] These contrasts confirm quite clearly the importance of freedom for human flourishing.

Equality

In a well-functioning society, there is a high degree of mutual respect between its members. Can such a situation be achieved if there are massive gaps in income between rich and poor in a society?

Evidence of the effects of income inequality on the evaluations of happiness is mixed. There is of course the basic point that in any society the value of an extra dollar to a poor person is much greater than to a rich person. As we have seen, if we compare a poor person with someone who is x times richer, an extra dollar is worth x times more (in terms of life satisfaction) to the poor person than to the one who is richer. So in a country with a given GDP per head, the average life satisfaction must logically (other things equal) be greater if the income is more equally distributed.[79]

On top of that mechanism (working through individuals), it is often said that inequality damages happiness through increased social tensions. For example, Wilkinson and Pickett claim that income inequality is associated with lower well-being of various kinds.[80] But they also find that inequality damages the rich as well as the poor – indicating some kind of environmental effect.

Nevertheless empirical work on the effects of inequality on life satisfaction has yielded very mixed results. Many studies have failed to find any effect.[81] The most positive results are in an interesting time-series study using both the U.S. General Social Survey and Eurobarometer.[82] This finds that in both the U.S. and Europe increases in inequality have (other things equal) produced reductions in happiness. The effect has been stronger in Europe than in the U.S. This difference probably reflects ideological differences: some 70% of Americans believe that the poor have a chance of escaping poverty, compared with only 40% of Europeans. Interestingly, the actual facts are actually the other way round: there is more intergenerational social mobility in Europe than the U.S. And there is more mobility where there is greater income equality.[83] But attitudes have an effect on perceptions and thus on happiness.

People hate inequality much more when they think it is unfair. For example, in some transition countries, particularly Poland, income inequality, initially perceived as a positive signal of increased opportunities, started to undermine people's life satisfaction when individuals became skeptical about the legitimacy of the enrichment of those who won out in the reform process.[84]

The conclusion must be that, other things equal, equality is desirable for two reasons. First the value of extra income is greater for the poor than the rich. And second, greater equality can be associated with reduced social tensions, especially when inequality is perceived as unfair. But greater equality is unlikely to come about without some greater pre-existing ethos of mutual respect and solidarity.

Values and Religion

This brings us directly to the issues of values, including those connected with religious belief. Clearly the values of a society are crucial to the inhabitants of a society. They are important in two obvious ways: a person's happiness depends on his own values but also on the values of those around him.

Many people get their values from religion but many do not. The overlap between values and religious belief has not yet been studied in the happiness literature, so we shall treat these as two separate issues, beginning with religion, before coming on to those values that can also be expressed in purely secular terms.

Religion

Some 68% of adults in the world say that "religion is important in their daily lives."[85] Yet our understanding of its effects on human happiness is limited. The Gallup World Poll data reported in Chapter 2 provide a starting point.[86] They show that religious belief and practice is more common in countries where life is harder (less income, life expectancy, education and personal safety). They also show that in the U.S. religious belief is higher in those states where life is harder. After controlling crudely for those factors, there is no difference in life satisfaction between more and less religious countries. There is however a clear difference when comparing the emotional life of more and less religious regions. In particular, in those countries where life is tough, there is strikingly more positive emotion and less negative emotion among those people who are more religious. Where life is easier, there is no such difference in this study.

It is interesting to understand what aspects of religion produce the positive effects on happiness. Clearly religion has both social aspects (especially through attending places of worship) but also deeply personal aspects (as connected for example with private prayer). In the Gallup World Poll people are asked about the importance of religion in their daily lives and also about whether they "have attended a place of worship or religious service within the last seven days?" (roughly half of the world's population had done this). Though these variables are not perfectly correlated they both have similar explanatory power.[87]

It is therefore natural that, when further questions are examined, they confirm that religion can help in hard circumstances both by providing more "relatives or friends you can count on," and more feelings of being respected, and more feeling that "your life has an important purpose or meaning."[88]

Most of the results we have considered above are based on inter-country comparisons. When it comes to comparisons between individuals there is always the problem that people who are naturally happier in given circumstances are more willing to believe that there is a benevolent deity. However studies of individuals do largely agree with the preceding inter-country findings. Meta-analysis concludes that greater religiosity is mildly associated with fewer depressive symptoms[89] and 75% of studies find at least some positive effect of religion on well-being.[90] This effect is particularly prevalent in high-loss situations, such as bereavement, and weaker in low-loss situations, such as job loss or marital problems. Thus religion can reduce the well-being consequences of stressful events, via its *stress-buffering* role.[91]

A recent large study of individuals in the European Social Survey found small but statistically significant effects on life satisfaction of "ever attending religious services" and "ever praying."[92] And interestingly the religiosity of others in the region was also found to have positive benefits both on those who are religious and on those who are not. This confirms findings from cross-country analysis of the Gallup World Poll that weekly church attendance has positive spillovers on the well-being of others at the national level.[93]

Altruism

But many of the values taught by the world's religions are of course universal values that have also been strongly supported by secular systems of ethics, going back to Stoicism (the most prominent ethical system of the Roman Empire) and beyond. The central principle is "do as you would be done by" – behave to others as you would wish them to behave to you. This frequently requires that you incur costs for the benefit of others – the fundamental definition of altruism.

Clearly altruistic behavior benefits those at the receiving end. But does it also benefit those who give, as well as those who receive? There is substantial evidence that it does, and that this is why it is so much more common than the crude teachings of elementary economics might predict.

There is of course plenty of evidence that people who care more about others are typically happier than those who care more about themselves.[94] But does that mean that altruism increases happiness in a causal sense? Evidence on volunteering and on giving money suggests that it does.[95]

When East Germany was united with West Germany, many opportunities for volunteering in East Germany disappeared. At the same time those who had previously volunteered experienced much larger falls in happiness than those who had not been volunteering. This suggests strongly that volunteering had been a cause of happiness for those who did it.[96] Acts of kindness have a similar effect – in a randomized experiment, the treatment group was told to do three extra acts of kindness a day and this significantly raised their happiness for some weeks.[97]

In an experiment on giving, one group was given some money to spend on themselves and another group was given equal amounts of money to spend on others. At the end of the day the second group reported themselves to be the happier.[98] These effects on happiness can also be observed in the brain's reward centers – when people give money they experience a positive reward.[99] Moreover altruism can be trained. After two weeks' compassion training, a control group gave more money in a laboratory game and showed more neural activity in the reward centers of the brain.[100]

There is a parallel question of whether happiness in turn increases altruism – a key question if we are wondering whether greater individual happiness would also increase happiness in others. There is some evidence that happiness is contagious.[101] But the specific channel of altruism is best studied through experimentation. A number of experiments have confirmed that happier people are more likely to give help to others.[102]

Materialism

Most ethical systems teach not only altruism but also that material wealth should not be pursued beyond the point where it compromises other values. Many studies have shown that other things equal, people who care more about money are less happy.[103] An important study covers a group of students who were freshmen in 1976.[104] Soon after entering college they were asked "the importance to you personally of being well off financially." Nineteen years later they reported their income and their overall satisfaction with life, as well as with family life, friendships, and work. At a given level of income, people who cared more about their income were less happy with life overall, with their family life, with their friendships and with their job. Of course people who care more about money also tend to earn more, and this helps to offset the negative effect of materialism. But in this study a person considering high income essential would need twice as much income to be as happy as someone considering high income unimportant.

If materialistic values tend to reduce social life, so does watching TV. Many studies have shown that watching TV is associated with lower happiness, other things equal. An early study exploited the fact that one Canadian town gained access to TV some years later than other towns.[105] The result was a relative fall in social life and increased aggression. So TV may cause problems in many ways, including reduced social life and increased violence. But the U.S. General Social Survey also shows that heavy TV watchers see so many rich people on the screen that they underestimate their own relative income.[106] Against all this TV also provides a great deal of enjoyment and instruction.

Environment

A final issue of values is the environment. There are two quite distinct issues here. One is the future of the planet. This mainly affects the happiness of future generations, rather than adults currently alive. The issue of greenhouse gases is a classic free-rider problem, and negotiation alone will not solve it. It will require a major dose of altruism world-wide.

The second environmental issue is the effect of our existing environment on adults who are currently alive. The environment we live in is extremely complex and only a few of its aspects have so far been examined for their effects on human happiness. The method is the same standard method we have used so far – to compare the happiness of people living in different environments.[107] Dimensions which have been examined so far include air quality (sulfur dioxide),[108] airport noise,[109] and aspects of the climate (sun, heat, humidity and wind).[110] In all cases sensible results have been obtained. Other studies have examined the effects of the natural world on human experience. At a very primary level, people assigned to walk from A to B on a tree-lined path alongside the Rideau River were systematically happier than those taking the same trip via the Carleton University's underground tunnel system, and the actual gains were much higher than people thought they would be.[111] Students who can see greenery out their classroom windows do better than those who cannot.[112] A hospital window with a green view similarly sees patients cured faster,[113] and there are many other studies linking green spaces to better health, performance, and life satisfaction.[114]

Mental Health

So far we have focused on factors that are very heavily influenced by the society in which you live. We turn now to factors that vary more within a country than across countries, but which are still extremely important areas for public policy.

Before starting, there is one important point to make. Happiness depends crucially on personality, and personality is strongly affected by your genetic make-up. This has been conclusively established by studying the similarity of happiness among identical twins reared apart, and comparing it with the similarity of happiness among non-identical twins who grew up in the same family. Identical twins reared apart are remarkably similar in their levels of happiness, while non-identical twins brought up together are very different.[115]

One important channel through which genes operate is mental health. This can be seen by taking any serious mental illness such as bipolar (or manic-depressive) disorder. If one identical twin has this condition, the other will also have it in 65% of cases. But, if the twins are non-identical, this falls to 14%.[116] For less serious mental health conditions the role of the genes is smaller. But for any condition, however serious, environmental interventions can also make a huge difference, as we shall see.

So how important is mental health in explaining the variation of happiness within any particular country? There is obviously a danger here of tautology – we explain the variation of happiness by the variation of misery. However, we can largely deal with this problem by measuring mental health earlier in life and using it to explain current happiness. We can illustrate this approach using the British Cohort Study of people born in 1970. For example, we can measure their life satisfaction when they were 34, and then explain this by the standard factors discussed in this chapter plus their level of malaise eight years earlier.[117] Of all the influences the most powerful were malaise eight years earlier ($\beta = -.23$), general health eight years earlier ($\beta = .10$) and current income ($\beta = .10$). Even if mental health is measured at age 16 it still exerts nearly as much impact on life satisfaction at age 34 as does current income. These facts in themselves have profound implications for policy.

These are the direct effects of mental health, holding constant a person's current circumstances. But mental health also has an indirect effect, through its effect on those current circumstances. To see this, we have to again look back at the effects of previous mental illness. About one half of those who are mentally ill as adults were already ill by the age of 15 (one half with primarily emotional problems and one half with primarily behavioral problems).[118] If we take those people who were mentally ill as adolescents, we can compare their adult circumstances with those of the rest of the population. Holding other things constant, they are more likely to have experienced low earnings, unemployment, criminal records, teenage pregnancy, physical illness and poor educational performance.[119] And these factors will in turn reduce their happiness – and frequently that of other members of the community as well.

Even if mental illness is quite narrowly defined, it affects a large number of people. The estimates in population surveys vary between countries for reasons that are not well understood.[120] But in the typical advanced country roughly 15% would be assessed as ill enough to need treatment – with some 1% suffering from psychotic conditions (especially schizophrenia) and most of the rest divided equally between depression on the one hand and on the other hand anxiety disorders like social phobia, panic attacks, obsessive-compulsive disorder, PTSD and crippling general anxiety. In developing countries rates of psychosis are similar, but measured rates of depression and anxiety are somewhat lower.

Mental illness is extremely disabling. For example a recent WHO study estimated that depression was 50% more disabling than chronic physical illnesses like angina, asthma, arthritis or diabetes.[121] If we combine the high prevalence of mental illness with its severity, we find that in WHO estimates it accounts for 43% of disability in advanced countries (as measured by the WHO) and 31% of disability world-wide.[122] Indeed among the working-age population in advanced countries, mental illness accounts for as much disability as all the other diseases put together. Even if we include all age groups and measure the overall burden of disease so as to include not only disability but also premature death, mental illness accounts for 26% of the burden of disease in advanced countries, and 13% world-wide.

Yet mental illness is in very many cases curable – more so than many physical diseases – and in nearly all cases it is treatable with significant benefit. For example, when people with anxiety conditions (which have often lasted for decades) are treated by cognitive behavioral therapy, one half will experience a complete and permanent recovery. Similarly one half of depressed patients will recover and have far less risk of relapse.[123] These treatments are not expensive. Medication is also an effective treatment for severe depression, with recovery rates of around one half and reduced risk of relapse if the medication is continued.

Despite this, in most advanced countries only a quarter of people with mental illness are in treatment, compared with well over three quarters for most physical conditions. This is a cause of much unnecessary misery, and the situation is even worse in developing countries. Though genes and childhood experience affect our likelihood of mental illness, healthcare interventions in childhood and later can reverse the course of our lives.

And increasingly preventive interventions are also being developed, which can reduce the likelihood of developing mental illness in the first place. These include interventions that can be made in childhood or pursued by adults. They all fall under the broad heading of "mind-training" and have been shown to affect not only self-reported well-being but also immune responses, hippocampal activity and educational performance.[124]

Physical Health

Physical health too is obviously correlated with well-being. There are however a number of issues involved in establishing how far health affects happiness. For example the most common measure of health used in these surveys is a subjective one. In the BHPS individuals are asked "Please think back over the last 12 months about how your health has been. Compared to people of your own age, would you say that your health has on the whole been…," with response categories of Excellent, Good, Fair, Poor, and Very Poor. While these kind of measures are easy to apply, and are indeed widely used, it is possible that individual replies are influenced by response style (some people give more positive replies, while others are more negative for the same level of underlying health), and that the same response bias is at work in each person's answers to the well-being questions. If so, it is unsurprising that health and well-being are correlated, but this need not necessarily reflect any causal relationship.

One way of dealing with this problem is to use panel data, in which we consider changes in health for the same individual and relate these to changes in lif- satisfaction. This method of analysis will eliminate any fixed individual response style with respect to either health or well-being. The data in the Appendix to this chapter enable us to adopt this approach. Table 1 in the Appendix shows that cross-section analysis of BHPS, GSOEP and WVS data produces estimated coefficients on self-assessed health that are not only significant, but also very sizeable: ceteris paribus, reporting health in the top two categories is associated with life satisfaction scores that are two to three points higher than reporting health in the worst category. Table 2 in the Appendix then turns to panel analysis for the two datasets in which individuals are repeatedly interviewed (BHPS and GSOEP). The estimated coefficients in the panel analysis are indeed somewhat smaller than those in Table 1, but nonetheless remain significant at all conventional levels. Thus health has a large impact on individual life-satisfaction.

Another approach is to avoid subjective health measures altogether and turn to more objective measures, such as doctor visits, nights spent in hospital, or measures of disability. There can be debate over whether such happenings are partly caused by underlying well-being, but these variables are at least more objective than self-reported health. There is also self-reported information from the British Household Panel Survey on whether the individual is moderately or severely physically disabled, and this can be related to life satisfaction.[125] Panel data allows us to compare the life-satisfaction of the same individuals before and after they became disabled. The impact effect of severe disability is estimated as being 0.6 points on the one to seven life satisfaction scale, and that of moderate disability as 0.4 points. There is also evidence of adaptation to disability, such that someone who has been disabled for all of the past three years is less affected (in life-satisfaction terms) than someone who is recently disabled. This adaptation is estimated at around 50% for moderate disability and 30% for severe disability (so that around one-third of the life-satisfaction effect of the latter dissipates over time).

There is also a reverse relationship: the impact of happiness on health. More happiness predicts better future physical health.[126] The medical literature has found high correlations between various low well-being scores and subsequent coronary heart disease,[127] strokes,[128] suicide[129] and length of life.[130] Individuals with higher positive affect have better neuroendocrine, inflammatory and cardiovascular activity.[131] Those with higher positive affect are less likely to catch a cold when exposed to a cold virus, and recover faster if they do.[132]

The Family

Marriage

Loving and being loved are key conditions for human happiness. Marriage is an institution which is meant to promote those experiences. Does it?

Marriage is one of the unambiguous, universally positive and statistically significant correlates of life satisfaction. Basic estimates of happiness always reveal that being married rather than single, divorced or widowed, is strongly associated with higher self-declared happiness, in all countries that have been under study, e.g. the United States and the countries of the European Union,[133] Switzerland,[134] Latin America, Russia, Eastern Europe[135] and Asia.[136] In most countries married people are also happier with their life than those who cohabit with a partner.

But does marriage make people happy, or are happy people more likely to become and remain married? An obvious way to try and disentangle the two directions of causality is to use longitudinal surveys and follow the same individuals over time. Accordingly, a retrospective view over 17 years of individuals living in Germany[137] reveals that among single people aged 20, those who would get married later were already happier with their life at the age of 20 than those who would remain single. Also, those who would get divorced were already less happy when they were single or newly married. Hence individuals who are already happier when they are young have a higher probability of becoming and remaining married. Even so, above and beyond these selection effects in and out of marriage, getting married still gives an additional boost to happiness, at least for some years.

Life satisfaction peaks in the years before and after marriage. As with many other life events, the happiness boost is subject to considerable adaptation after a honeymoon period. But for those who never get divorced, happiness remains permanently higher than before they were married.[138]

To many readers this marriage “bonus” will not come as a surprise. Apart from love and companionship, there are economic advantages of marriage, such as insurance and buffers against adverse life shocks, economies of scale and specialization within the family;[139] and studies have documented the fact that, compared to single people, married people enjoy better physical and psychological health (e.g. less substance abuse and less depression) and live longer.[140] Because marriage involves a long-term commitment, accompanied with trust and companionship, it can also be seen as a form of social capital. Research where the data permit has shown that the happiness of spouses is interdependent, in Great Britain,[141] Germany[142] and Australia.[143]

However, as for any social group, couples may also suffer from inter-personal comparisons and relative deprivation. The leisure and social activities of one's spouse cause reduced life satisfaction in the other spouse.[144] Intra-household comparisons of happiness can be a cause of marital instability:[145] couples with a higher happiness gap are more at risk of divorcing in the future, especially if the wife is the unhappy one.[146] A related observation is that assortative mating (i.e. marriage between equally educated people) seems to favor greater happiness and lower risk of divorce,[147] probably because there is more similarity and experience-sharing between spouses.

Finally, should people who are unhappy in their marriages expect to be happier after divorce? It is true that divorced people are less happy and suffer from more mental strain than people who remain married; but people who live in a marriage of (self-assessed) poor quality are less happy than unmarried people.[148] As British, German and American data show, a person who gets divorced becomes on average more satisfied with their life, some years after the separation, than he or she used to be in the three years preceding separation.[149]

Of course the break-up of a couple is also likely to affect other people, and children are obvious potential collateral victims. Evidence of the impact of parental separation on children's welfare is abundant; poor school performance being one of the most obvious symptoms. However, it is not divorce as such that appears to damage children's welfare and school grades, but parental conflict. Accordingly, children's school performance deteriorates for several years before the official separation of their parents.[150] However, the issue of child development is too large to be treated here and most of the evidence comes from quite different sources from those considered in this report.

In sum, (a good) marriage is a source of life satisfaction, and conversely, the equality of happiness between spouses is a guarantee of marital stability; less happy people are more likely to get divorced, but once they do, divorcees reach higher levels of happiness in the long run than they used to experience before divorce.

Children

But do children make their parents happy? Surprisingly, the presence of children in the household appears not to be associated with higher life satisfaction.[151] This is found in the World Values Survey and in panel data for U.K. and Germany (see Appendix). Several surveys of the literature acknowledge this surprising absence of a relationship[152] and in particular single parents with more children are less happy than those with fewer children.[153]

There are obviously the responsibilities and time pressures of childcare, especially for those facing too many claims on their time. Time-use and experience-sampling studies, which investigate the occurrence of positive and negative feelings (happy, relaxed, frustrated, depressed, angry, sad, etc.) during typical daily activities, reveal that childcare ranks very low in the hierarchy of daily activities, sometimes as low as 16th out of 19 daily activities, in terms of net positive feelings.[155] Everyday experience with children may not be immediately rewarding, but "*a man who has children lives like a dog, a man without children dies like a dog*," says the proverb. However the effect of adult children has not yet been systematically investigated. Moreover even among young children, their age matters. Young children under 3 and teenagers are associated to a lower level of parents' happiness, whereas children aged 3-12 are associated with higher happiness.[155]

Richer people are on average happier with being parents, and parenthood is also less problematic in the social-democratic countries of Northern Europe, where there is more child-support from the state.

In summary, having children is no guarantee of higher happiness. The pleasure of parenting depends on the age of the children, on the quality of the parenting couple and on the social context, including having enough time to enjoy family life.

Education

On average, the level of education has no clear direct impact on happiness, but education is of course indirectly related to happiness through its effect on income: education increases income and income increases happiness. Studies of the financial returns to education in recent years, mostly based on natural experiments, show high returns from additional years of schooling (between 7% and 15% per year).[156] Longer years of education are also associated with increased employability and job security, and faster promotion,[157] all of those being factors conducive to higher happiness.

By contrast, evidence on the direct effect of education is mixed and varies between countries.[158] One obvious problem is that happy people may be more likely to persist in education and this effect cannot be controlled for in panel studies. But there is one type of natural experiment which can help – the raising of the compulsory minimum school leaving age. This has been shown to have directly raised the average happiness of those affected by the change, though again largely through its effect on income.[159]

The conclusion is that education may have some non-income benefits to the individuals who get an education, especially in poor countries.[160] But this is smaller than is often claimed by educationalists. On top of that there may be important social effects through an informed electorate and in poor countries through reduced birth-rates and mortality.

Gender

In most advanced countries women report higher satisfaction and happiness than men.[161] In our Appendix, women report higher life satisfaction scores than do men in all three of the data sets analyzed. But this finding is dominated by advanced countries. Outside the industrial countries the happiness gap infavor of women is often found to be smaller or even reversed.[162] Moreover in both the U.S. and Europe women are becoming less happy relative to men.[163]

In the U.S. it is also possible to investigate gender differences using the U-index, which is defined as the proportion of time spent in activities for which the highest-rated feeling was negative.[164] Data from the Princeton Affect and Time Survey (PATS), where the activities of the day previous to the interview are reconstructed, show that U.S. women have lower U-index scores than men – and thus less misery.[165] It is also found that women are relatively happier in countries where gender rights are more equal.[166]

Though women report higher life satisfaction than men, ceteris paribus, their rates of mental illness are also higher.[167] In the BHPS, where both well-being measures are available, women report higher overall life satisfaction scores but more psychological stress, as measured by the twelve-item GHQ-12.[168]

Age

The relationship between age and evaluations of happiness is one of the most robust and common findings in happiness research. *A priori*, most people would expect that happiness steadily declines with age, at least in adulthood, as do many of our physical and mental faculties. But the pattern of life-evaluation uncovered by surveys is essentially U-shaped through life: satisfaction declines, reaches a minimum in middle-age (between 40 and 50), and then rises again. This two-part age profile of life-evaluation has been observed in many countries in many continents,[169] although some work finds different patterns in specific countries.[170] In the U.S. the U-shaped pattern has also been found for affect.[171]

The shape is even more pronounced when holding income, marital status, health and employment status constant. Thus it is not only higher income or greater family stability that explains the rebound in happiness after the mid-life low. Explanations could include the wisdom of maturity, or the beneficial effect of reduced (or more realistic) aspirations. However between 70 and 80 worsening health begins to take its effect[172] and average happiness begins to decline once more.[173]

Conclusion

We have covered a lot of ground. We have shown above all that happiness depends on a huge range of influences, many of which can be influenced by government policy. In due course we shall have more precise estimates of these effects. But a very rough perspective of orders of magnitude can be got from Appendix B, where we compare the effects of the main influences looked at in this chapter with the effect of a 30% increase in income. The calculations confirm the powerful effect of many variables other than income.

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1 We are extremely grateful to the U.S. National Institute of Aging (R01AG040640) for financial support, to John Berry, Angus Deaton, Ed Diener, Bruno Frey, Carol Graham, Richard Easterlin, John Helliwell, Daniel Sachs, Andrew Steptoe, Betsey Stevenson and Alois Stutzer for helpful comments and to Harriet Ogborn for expert handling of the preparation of the manuscript.

2 On the role of genes see Lykken (1999) and Plomin et al. (2001).

3 However on the other side, estimates of effects based on longitudinal data are more biased towards zero by measurement error. They also reflect a narrower range of variation of the variables under study.

4 For a classic review of the determinants of happiness see Diener & Biswas-Diener (2008). See also Dolan et al. (2008).

5 Layard et al. (2008). This implies no level of income at which people become satiated. Kahneman & Deaton (2010) have examined this question in the U.S. for a given year. Holding constant the incomes of other people, individual life-evaluation has no point of satiation but positive affect becomes satiated with income around the level of $75,000.

6 Layard et al. (2010), Fig 6.2.

7 Layard et al. (2010), Table 6.1.

8 See Appendix A. A similar reduction is found using the British Household Panel Survey. This reduction reflects two features: the introduction of the fixed effect and the greater measurement error bias in panel data. Bound & Krueger (1991) suggest that for income the measurement error reduces the coefficient estimate by a factor of around 0.8 in the cross-section and 0.65 in first differences.

9 Layard et al. (2010), Table 6.4.

10 Of course in any year richer people have on average more transitory income. But they also have more permanent income.

11 Di Tella et al. (2010) find strong effects of lagged income but this is because they exclude comparator income from their explanatory variables (and include occupational status instead).

12 Clark et al. (2008b).

13 For experimental evidence of such effects see Oswald et al. (2011). But there have been good attempts to handle this problem using industry wages or twins as instrumental variables. See Pischke (2011) and Li et al. (2011).

14 Knight et al. (2010). However in South Africa, Kingdon & Knight (2006) find that the comparator incomes that have negative effects on happiness are mainly from outside the village.

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[15] Clark & Senik (2010).

[16] Mayraz et al. (2010).

[17] Layard et al. (2010).

[18] For studies which find effects in the U.S. and Canada, see Luttmer (2005), Helliwell & Huang (2010), and Barrington-Leigh & Helliwell (2008).

[19] Hirschman & Rothschild (1973)

[20] Graham & Pettinato (2002).

[21] In individual regressions in the Gallup World Poll, individual income raises life satisfaction as usual but GDP (the comparator) does not reduce it (Helliwell et al., 2010).

22. Dohmen et al. (2011)

23. See also Deaton (2008).

24. *Source*: Gallup World Poll 2005-2011, equations by Helliwell and Wang. The measure of life evaluation is the ladder. Very similar results would be obtained using life satisfaction, see Helliwell et al. (2010), Table 1, for a comparison using identical samples.

25. Stevenson & Wolfers (2008b).

26. The U.K. is another example.

27. See Easterlin et al. (2010).

28. Sacks et al. (2012). Scores for each year indicate the average value in that year for all countries covered. The units are Z-scores net of country FEs. The table implies that over the last 40 years average happiness has risen in total by 0.14 standard deviations of happiness across the world's population.

29. Sacks et al. (2012), Table 3, Lines 5 and 6. Also, in this study changes in happiness are only measured using the same survey source for every observation.

30. One source of controversy relates to the maintained hypothesis. For example, in many estimates the effect of economic growth on life satisfaction is not significantly different from zero, nor is it significantly different from the cross-sectional effect across countries or individuals.

31. Di Tella et al. (2003). Helliwell & Huang (2011b) use cross-sectional U.S. evidence to show that well-being is higher in places where unemployment is lower.

32. Kahneman (2011). See also Wolfers (2003) who shows the harmful effects of unemployment variability.

33. See, for example, Blanchflower & Oswald (2004), Clark & Oswald (1994), and Winkelmann & Winkelmann (1998).

34. Eisenberg & Lazarsfeld (1938).

35. Jahoda (1981 and 1988).

36. McKee & Bell (1986).

37. Green (2011).

38. Luechinger et al. (2010).

39. Helliwell & Huang (2011b).

40. Jackson & Warr (1987).

41. Clark (2003). This finding is confirmed for large samples of the U.S. data in Helliwell & Huang (2011b).

42. Platt & Kreitman (1990) and Platt et al. (1992).

43. Grün et al. (2010), p.287.

44. Wulfgramm (2011). See also Gyarmati et al. (2008) for a randomized experiment in which the treatment group were allocated to work on community projects.

45. See Warr (1999) for example.

46. Clark (2010).

47. Deci & Ryan (1985).

48. Benjamin et al. (2012).

49. Ariely et al. (2005), Pink (2009), Frey (1997).

50. Kahneman et al. (2004).

51. Robertson & Cooper (2011) and Cooper & Lundberg (2011).

52. Blanchflower & Oswald (1998), Blanchflower et al. (2001) and OECD (2000).

53. Frey & Benz (2008).

54. Bianchi (2012).

55. Ibid.

56. Frey (2008), Chapter 7.

57. Graham & Pettinato (2002).

58. Clark & Fawaz (2009). Many studies find no overall effect of retirement on health (Coe & Lindeboom, 2008), but some find a negative effect of retirement on mental health (Dave et al., 2008). In general this literature has not produced an unambiguous set of results.

59. Maslow (1943).

60. The OECD (2001, p. 41) defines social capital as "networks together with shared norms, values and understandings that facilitate cooperation within or among groups."

61. Knack (2001).

62. Helliwell & Putnam (2004); Helliwell (2008); Helliwell & Wang (2011a).

63. This dominance of workplace trust applies both for trust in colleagues (Helliwell & Wang 2011a, and Helliwell & Barrington-Leigh 2011) and trust in management (Helliwell & Huang 2010). These two measures of workplace trust are compared, and their pre-eminence confirmed, in Helliwell & Huang (2011a).

64. Powdthavee (2008).

65. Meier & Stutzer (2008).

66. Algan & Cahuc (2010), Uslaner (2008).

67. Soroka et al. (2007), Helliwell & Wang (2011a)

68. Durante (2009).

69. Nunn & Wantchekon (2011).

70. Bartolini (2011), Sarracino (2010).

71. Putnam (2000).

72. Bowles & Gintis (2002). Of course, "*communities work because they are good at enforcing norms, and whether this is a good thing depends on what the norms are*" (p.428). Indisputably, examples of harmful norms, such as the culture of honor and the associated violence, are legion. See Nisbett & Cohen (1996); Grosjean (2011).

73. Helliwell & Wang (2011a).

74. Alesina & La Ferrara (2005). See also Sampson et al. (1997).

75. See Nguyen & Benet-Martinez (forthcoming) for a meta-analysis of biculturalism.

76. Stevenson & Wolfers (2008a).

77. Singer (1981), Pinker (2011).

78. See also Easterlin (2010).

79. Stevenson & Wolfers (2010) use cross-country data to examine this mechanism and cannot refute it (nor can they refute zero effect of inequality).

80. Wilkinson & Pickett (2009). They do not include happiness or life satisfaction in their outcomes.

81. Stevenson & Wolfers (2010), Blanchflower & Oswald (2004) and Helliwell (2003), p.351. But for positive results see Morawetz et al. (1977) and Schwarze & Härpfer (2007).

82. Alesina et al. (2004).

83. Blanden (2009), Atkinson et al. (1992), Burkhauser & Poupore (1997).

84. Grosfeld & Senik (2010).

85. In this context Buddhists normally report themselves as religious, even if others question this use of words.

86. Diener et al. (2011), see especially their Figure 1. This also shows that there was a large difference in happiness between religious and less religious the U.S. states but only a small one for the U.S. individuals.

87. See *op cit*, Table 3.

88. By contrast, recent theU.S. research has found evidence suggesting that all of the life satisfaction benefits of religiosity flow through church attendance and the social networks provided by their congregations, and specifically friendships among those sharing a strong sense of religious identity, see Lim & Putnam (2010). Another ingenious the U.S. study uses exogenous changes in laws related to Sunday shopping to identify a positive effect of church attendance (Cohen-Zada & Sander, (2010)).

89. Smith et al. (2003).

90. Pargament (2002).

91. Ellison (1991).

92. Clark & Lelkes (2009).

93. Helliwell et al. (2010).

94. Lyubomirsky (2007). See also Singer et al. (2004).

95. See also Brown et al. (2003) and Schwartz & Sendor (1999).

96. Meier & Stutzer (2008).

97. Lyubomirsky (2007), Chapter 5.

98. Anik et al. (2010). See also Aknin et al. (2010) and Dunn et al. (2008). In the cross-country regressions in Table 3.1, it is possible to add a further variable: the proportion of subjects who gave money to a charity in the last month. This is again highly significant.

99. Harbaugh et al. (2007). See also Zaki & Mitchell (2011) who also show that inequitable behavior causes activity in brain regions as sociated with subjective disutility.

100. Experiment conducted by Richard Davidson.

101. Fowler & Christakis (2008).

102. Anik et al. (2010).

103. Konow & Earley (2008).

104. Nickerson et al. (2003).

105. Williams (1986).

106. Layard (2005), Annex 6.1. See also Bruni & Stanca (2008).

107. In principle we need to allow for the fact that people have chosen their environments.

108. Luechinger (2009).

109. Van Praag & Baarsma (2005).

110. Frijters & Van Praag (1998), and Brereton et al. (2008), who also examine proximity to the sea, to landfill, and to transport routes.

111. Nisbet & Zelenski (2011).

112. Matsuoka (2010).

113. Ulrich (1984).

114. Basu et al. (2012).

115. Lykken (1999).

116. McGue & Bouchard (1998), Table 1.

117. Layard (2012). Similar findings come from the earlier National Child Development Study of people born in 1958.

118. Kim-Cohen et al. (2003).

119. Richards & Abbott (2009), Layard & Dunn (2009).

120. Demyttenaere et al. (2004).

121. Moussavi et al. (2007).

122. WHO (2008).

123. Roth & Fonagy (2005).

124. Durlak et al. (2011); Williams et al. (2007); Davidson et al. (2003) and Hölzel et al. (2011).

125. Oswald & Powdthavee (2008).

126. See Diener & Chan (2011), Frey (2011) and Lyubomirsky et al. (2005).

127. Sales & House (1971).

128. Huppert (2006).

129. Koivumaa-Honkanen et al. (2001).

130. Palmore (1969) and Mroczek & Spiro (2006).

131. Steptoe et al. (2005).

132. Cohen et al. (2003).

133. Di Tella et al. (2003).

134. Frey & Stutzer (1999).

[135] Graham & Pettinato (2002); Senik (2004); Grosjean et al. (2011).

[136] Knight et al. (2010); Knight & Gunatilaka (2010).

[137] Stutzer & Frey (2006).

[138] Stutzer & Frey (2006).

[139] Becker (1974).

[140] Gardner & Oswald (2002); Gove et al. (1983); Hu & Goldman (1990).

[141] Powdthavee (2009).

[142] Lucas & Schimmack (2006).

[143] Guven et al. (2012, forthcoming).

[144] Clark (2011).

[145] Guven et al. (2012, forthcoming).

[146] Ibid.

[147] Frey & Stutzer (2006).

[148] Chapman & Guven (2010).

[149] Gardner & Oswald (2006); Clark et al. (2008a).

[150] Piketty (2003).

[151] Di Tella et al. (2003); Alesina et al. (2004).

[152] Blanchflower (2008); Clark et al. (2008a); Layard (2005); Gilbert (2006). But few studies find a positive association, see Clark & Oswald (2002); Frey & Stutzer (2006); Haller & Hadler (2006).

[153] Frey & Stutzer (2006).

[154] Kahneman et al. (2004).

[155] Margolis & Myrskylä (2011). Children affect happiness more favourably at weekends when parents have more time (Helliwell & Wang, 2011b).

[156] Angrist & Krueger (1991); Acemoglu & Angrist (2001); Card (2001).

[157] Blanchflower & Oswald (2004).

[158] For some positive results, see Frey & Stutzer (2002); Di Tella et al. (2003) and Appendix A for GSOEP and WVS but not BHPS. For evidence varying between countries, see Hartog & Oosterbeek (1998), p247.

[159] Oreopoulos & Salvanes (2011) relates to U.K., Canada and U.S.

[160] Hartog & Oesterbeek (1998) p. 247.

[161] See Blanchflower (2008) for evidence from repeated cross-sections covering a wide variety of countries, and Helliwell (2008) for Gallup data in 27 OECD countries.

[162] See Senik (2004) for instance. But no difference is observed in the Gallup World Poll.

[163] Stevenson & Wolfers (2009).

[164] Kahneman & Krueger (2006).

[165] Krueger et al. (2009).

[166] Graham & Chattopadhyay (2011).

[167] Nolen-Hoeksema & Rusting (1999).

[168] This BHPS measure is used in Clark & Oswald (1994) and Clark (2003).

[169] Blanchflower & Oswald (2004 and 2008); Hayo & Seifert (2003); Helliwell (2003); Helliwell et al. (2009).

[170] For the U.S. Easterlin (2001) has traced the same birth cohorts over their lifetime and failed to find the U-shape. If cross-section and cohort results differ, this must mean that the cohorts born some 40-50 years before the study had particularly bad lifetimes.

[171] Stone et al. (2010).

[172] Gerstorf et al. (2008); Gerstorf et al. (2010).

[173] Frijters & Beatton (2008); Wunder et al. (2009). This can also be seen in the GSOEP and the BHPS, using 10-year age groups. (This is despite the fact that the BHPS question refers to health relative to your age-group.)

Appendix A

Standard life-satisfaction equations for individuals in three large data sets

In this Appendix we present standard equations to explain life-satisfaction. We use the German Socio-Economic Panel (GSOEP, 1984-2009), the British Household Panel Study (BHPS, 1996-2008) and the World Values Survey (WVS, 1981-2008).

Table 1 shows OLS cross-section regressions, while Table 2 shows OLS equations including a fixed-effect for each individual and year, so that the equation estimates the effect of each variable in explaining the different levels of happiness which an individual experiences in each different year.

The sample used is prime-age (30-55), with the top and bottom 5% of income recipients trimmed. Income is household income. Regressions by Sarah Flèche.

Table 1: Cross-sectional regressions to explain life satisfaction
(Range of life satisfaction: GSOEP 0-10, BHPS 1-7, WVS 1-10)

	GSOEP (West)	BHPS	WVS
Log of Income (monthly)	0.60*(0.01)	0.26*(0.02)	0.65*(0.01)
Female	0.12* (0.00)	0.09*(0.01)	0.14* (0.01)
Age	0.11*(0.07)	0.11 (0.09)	0.19* (0.11)
Ageˆ2/1000	-3.55*(1.80)	-4.29* (2.18)	-5.37* (2.73)
Ageˆ3/1000	0.03*(0.01)	0.04* (0.01)	0.04* (0.02)
Single	-0.15*(0.01)	-0.34*(0.01)	-0.32* (0.03)
Widowed	-0.18* (0.04)	-0.46*(0.04)	-0.25* (0.03)
Divorced	-0.20* (0.01)	-0.41*(0.01)	-0.40*(0.03)
Separated	-0.48*(0.02)	-0.60*(0.03)	-0.55* (0.04)
Unemployed	-0.63*(0.01)	-0.33*(0.03)	-0.43* (0.02)
Self Employed	-0.15*(0.01)	0.07*(0.01)	0.04* (0.02)
Out of the labor force	-0.03* (0.01)	-0.13*(0.01)	0.11* (0.01)
Student	-0.12*(0.06)	-0.00 (0.06)	-0.34* (0.06)
Education: high	0.00* (0.01)	-0.20 (0.08)	0.14*(0.02)
Education: medium	-0.01(0.01)	0.00* (0.04)	0.07*(0.01)
One child	-0.02* (0.01)	-0.07*(0.01)	-0.03 (0.02)
Two children	-0.03* (0.01)	-0.06 (0.01)	-0.03 (0.02)
Three + children	-0.09*(0.01)	-0.09*(0.02)	-0.00 (0.02)
Health			
Excellent	3.45*(0.03)	1.94*(0.03)	2.69* (0.08)
Good	2.82*(0.03)	1.59*(0.03)	2.06* (0.08)
Satisfactory	2.04*(0.03)	1.09*(0.03)	1.44* (0.08)
Poor	1.26* (0.03)	0.59*(0.03)	0.61* (0.09)
Fixed effects	No	No	No
Time Dummies	Yes	Yes	Yes
Region Dummies	Yes	Yes	Yes
Observations	100,945	53,615	106,504
R^2	0.25	0.17	0.28

Note: * denotes statistical significance at 5% level

Table 2: Fixed effects regressions to explain life satisfaction
(Range of life satisfaction: GSOEP 0-10, BHPS 1-7)

	GSOEP (West)	BHPS
Log of Income (monthly)	0.39* (0.01)	0.13* (0.03)
Female	--	--
Age	-0.16* (0.07)	-0.08 (0.09)
Age^2 /1000	2.97 (1.76)	0.56 (2.13)
Age^3/1000	-0.02* (0.01)	0.00 (0.01)
Single	-0.07* (0.03)	-0.13*(0.04)
Widowed	-0.44* (0.07)	-0.18* (0.07)
Divorced	0.03 (0.02)	-0.14* (0.03)
Separated	-0.25* (0.03)	-0.34* (0.03)
Unemployed	-0.49* (0.01)	-0.22* (0.03)
Self Employed	-0.01 (0.02)	0.01 (0.02)
Out of the labor force	-0.13* (0.01)	-0.12* (0.02)
Student	-0.14*(0.06)	0.07 (0.06)
Education: High	0.07 (0.05)	-0.07 (0.08)
Education: Medium	0.10* (0.03)	0.13* (0.04)
One child	0.07* (0.01)	0.01 (0.01)
Two children	0.04* (0.02)	0.03 (0.02)
Three + children	0.06* (0.02)	0.06* (0.03)
Health		
Excellent	2.25* (0.03)	1.04* (0.03)
Good	1.92* (0.03)	0.90* (0.03)
Satisfactory	1.51* (0.03)	0.64* (0.03)
Poor	0.93* (0.03)	0.36* (0.03)
Fixed effects	Yes	Yes
Time Dummies	Yes	Yes
Region Dummies	Yes	Yes
Observations	100945	53615
R^2	0.20	0.09

Note: * denotes statistical significance at 5% level

GSOEP (1984-2009) (West Germany) Summary Statistics

Variable	Mean	Std. Dev	Min	Max
Life satisfaction	7.05	1.72	0	10
Female	0.51	0.49	0	1
Age	42.05	7.27	30	55
Age^2/1000	1.82	0.61	0.90	3.02
Age^3/1000	81.10	40.43	27	166.37
Single	0.13	0.34	0	1
Widowed	0.01	0.10	0	1
Divorced	0.08	0.28	0	1
Separated	0.02	0.16	0	1
Unemployed	0.06	0.24	0	1
Self-employed	0.05	0.22	0	1
Out of the labor force	0.12	0.33	0	1
Student	0.00	0.07	0	1
Log of Income	3.33	0.20	2.70	3.74
Education: high	0.19	0.39	0	1
Education: medium	0.31	0.46	0	1
One child	0.23	0.42	0	1
Two children	0.22	0.41	0	1
Three + children	0.08	0.27	0	1
Heath: excellent	0.09	0.29	0	1
Health: good	0.46	0.49	0	1
Health: satisfactory	0.30	0.46	0	1
Health: poor	0.10	0.30	0	1

BHPS (1996-2008) **Summary Statistics**

Variable	Mean	Std. Dev	Min	Max
Life satisfaction	5.07	1.24	1	7
Female	0.54	0.49	0	1
Age	42.52	7.12	30	55
Age^2/1000	1.85	0.61	0.90	3.02
Age^3/1000	83.41	40.35	27	166.37
Single	0.09	0.29	0	1
Widowed	0.01	0.10	0	1
Divorced	0.07	0.26	0	1
Separated	0.02	0.16	0	1
Unemployed	0.02	0.16	0	1
Out of the labor force	0.15	0.36	0	1
Student	0.00	0.07	0	1
Self-employed	0.09	0.29	0	1
Log of Income	3.40	0.22	2.80	3.79
Education: high	0.00	0.05	0	1
Education: medium	0.01	0.10	0	1
One child	0.22	0.41	0	1
Two children	0.19	0.39	0	1
Three +children	0.07	0.27	0	1
Heath: excellent	0.24	0.43	0	1
Health: good	0.47	0.49	0	1
Health: satisfactory	0.18	0.39	0	1
Health: poor	0.07	0.25	0	1

WVS (1981-2008) : 86 Countries Summary Statistics

Variable	Mean	Std. Dev	Min	Max
Life satisfaction	6.33	2.50	1	10
Female	0.51	0.49	0	1
Age	41.05	7.27	30	55
Age^2/1000	1.73	0.61	0.9	3.02
Age^3/1000	75.83	39.60	27	166.37
Single	0.09	0.28	0	1
Widowed	0.02	0.17	0	1
Divorced	0.04	0.19	0	1
Separated	0.02	0.14	0	1
Unemployed	0.07	0.27	0	1
Out of the labor force	0.20	0.40	0	1
Self-employed	0.14	0.34	0	1
Student	0.01	0.10	0	1
Log of Income	1.38	0.61	0	2.30
Education: high	0.22	0.41	0	1
Education: medium	0.42	0.49	0	1
One child	0.16	0.36	0	1
Two children	0.32	0.46	0	1
Three + children	0.38	0.48	0	1
Heath: excellent	0.21	0.41	0	1
Health: good	0.44	0.49	0	1
Health: satisfactory	0.27	0.44	0	1
Health: poor	0.05	0.23	0	1

Appendix B

Comparing the impact of different influences on happiness

For policy analysis it is necessary to compare the quantitative impact of different factors upon happiness. A natural way to do this is to compare the effect of a change in each other factor with the effect of a 30% increase in income.

At present the calculations are purely illustrative drawing on a variety of sources, which are indicated in the table. They do however show clearly how much else matters to people besides income.

Effects on life evaluation of each factor, as a multiple of the effect of a 30% increase in income

		Source
Individual unemployment (versus employment)	-6.0	*GSOEP Panel (Appendix A)*
Unemployment rate (10% point increase)	-1.3	*Helliwell and Huang, 2011b*
Social support (10% point extra saying yes)	4.5	*Table 3.1, unstandardized*
Freedom (10% point extra saying yes)	2.1	" "
Corruption (10% point extra saying yes)	1.9	" "
Malaise 8 years earlier (1 standard deviation worse)	-10	*British Cohort Study*
Physical health (Poor versus good, self-assessed)	-15	*GSOEP panel (Appendix A)*
Single (versus married)	-2.2	" "
Separated (versus married)	-4.0	" "
Widowed (versus married)	-2.9	" "
Female (versus male)	1.5	*GSOEP, using panel for effect of income*

Note: The effect of each other variable is compared with that of individual relative income. The exceptions are for social support, freedom and corruption, whose effects are compared with the effects of GDP per head.